Predicting Patterns in Sustainability and Development

Dataset Source: Kaggle
https://www.kaggle.com/datasets/anshtanwar/global-data-on-sustainable-energy

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Final Project - Exploratory Data Analysis
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I. Introduction

The global energy landscape has undergone major changes in the last 25 years fueled by needs to address climate change and energy security. Renewable energy sources such as solar, wind and hydropower have emerged as a solution to limit CO_2 emissions and to move away from fossil fuel energy. However, much of the energy resources used around the globe still come from non-renewable sources. As noted, "the global energy demand is projected to experience more increases in the coming decades, resulting in rising carbon emissions and creating global tension on how to deal with this growth" (Alzohbi, 2024).

Proper analysis on energy data is important as it can help policymakers make informed decisions. Predictive analytics, for example, "leverages historical solar irradiance data and machine learning models to improve short-term and long-term solar energy predictions, leading to better grid management" (Udo, 2023).

Through this paper, we will analyze the global energy and development data from 2000-2020 in order to Identify patterns, correlations, and anomalies. By leveraging machine learning techniques, including regression, classification and clustering, we aim to uncover hidden trends and relationships within the data. These methods enable us to predict future trajectories in energy consumption, renewable energy adoption and their impact on the economical development around the world.

II. Dataset Description

For the purpose of this analysis, we chose a dataset from Kaggle (https://www.kaggle.com/datasets/anshtanwar/global-data-on-sustainable-energy). It contains worldwide energy data spanning from 2000 to 2020, including energy access, sustainability indicators and economical indicators. The data is made up of 3649 entries and 20 features, which are summarized below:

- **Entity**: The name of the country or region for which the data is reported.
- **Year**: The year for which the data is reported, ranging from 2000 to 2020.
- **Access to electricity (% of population)**: The percentage of population with access to electricity.
- **Access to clean fuels for cooking (% of population)**: The percentage of the population with primary reliance on clean fuels.
- **Renewable-electricity-generating-capacity-per-capita**: Installed Renewable energy capacity per person
- **Financial flows to developing countries (US $)**: Aid and assistance from developed countries for clean energy projects.
- **Renewable energy share in total final energy consumption (%)**: Percentage of renewable energy in final energy consumption.
- **Electricity from fossil fuels (TWh)**: Electricity generated from fossil fuels (coal, oil, gas) in terawatt-hours.
- **Electricity from nuclear (TWh)**: Electricity generated from nuclear power in terawatt-hours.
- **Electricity from renewables (TWh)**: Electricity generated from renewable sources (hydro, solar, wind, etc.) in terawatt-hours.
- **Low-carbon electricity (% electricity)**: Percentage of electricity from low-carbon sources (nuclear and renewables).
- **Primary energy consumption per capita (kWh/person)**: Energy consumption per person in kilowatt-hours.
- **Energy intensity level of primary energy (MJ/$2011 PPP GDP)**: Energy use per unit of GDP at purchasing power parity.
- **Value_co2_emissions (metric tons per capita)**: Carbon dioxide emissions per person in metric tons.
- **Renewables (% equivalent primary energy)**: Equivalent primary energy that is derived from renewable sources.
- **GDP growth (annual %)**: Annual GDP growth rate based on constant local currency.
- **GDP per capita**: Gross domestic product per person.
- **Density (P/Km2)**: Population density in persons per square kilometer.
- **Land Area (Km2)**: Total land area in square kilometers.
- **Latitude**: Latitude of the country's centroid in decimal degrees.
- **Longitude**: Longitude of the country's centroid in decimal degrees.

	Year	Access to Electricity (%)	Access to Clean Fuels (%)	Renewable Elec. Capacity (per capita)	Financial Flows (US$)	RE Share Final Energy (%)
count	3649	3639	3480	2718	1560	3455
mean	2010.04	78.9337	63.2553	113.1375	9.42E+13	32.6382
std	6.0542	30.2755	39.0437	244.1673	2.98E+14	29.8949
min	2000.0	1.2523	0	0	0	0
25%	2005.0	59.8009	23.1750	3.5400	2.60E+11	6.5150
50%	2010.0	98.3616	83.1500	32.9100	5.67E+12	23.3000
75%	2015.0	100	100	112.2100	5.53E+13	55.2450
max	2020.0	100	100	3060.1900	5.20E+15	96.0400

	Fossil Fuel Elec. (TWh)	Nuclear Elec. (TWh)	Renewables Elec. (TWh)	Low-Carbon Elec. (%)	Primary Energy/Capita (kWh)	Energy Intensity (MJ/$PPP)
count	3628	3523	3628	3607	3649	3442
mean	70.3650	13.4502	23.9680	36.8012	25743.9817	5.3073
std	348.0519	73.0066	104.4311	34.3149	34773.2214	3.5320
min	0	0	0	0	0	0.1100
25%	0.2900	0	0.0400	2.8778	3116.7373	3.1700
50%	2.9700	0	1.4700	27.8651	13120.5700	4.3000
75%	26.8375	0	9.6000	64.4038	33892.7800	6.0275
max	5184.1300	809.4100	2184.9400	100	262585.7000	32.5700

	CO_2 Emissions (kt)	Renewables (% Eq. Primary)	GDP Growth (%)	GDP/Capita	Land Area (km²)	Latitude	Longitude
count	3221	1512	3332	3367	3648	3648	3648
mean	1.60E+11	11.9867	3.4416	13283.7743	6.33E+11	18.2463	14.8225
std	7.74E+11	14.9946	5.6867	19709.8667	1.59E+12	24.1592	66.3481
min	10	0	-62.0759	111.9272	21	-40.9005	-175.1982
25%	2020	2.1371	1.3833	1337.8134	25713	3.2027	-11.7798
50%	10500	6.2908	3.5599	4578.6332	117600	17.1898	19.1451
75%	60580	16.8416	5.8301	15768.6154	513120	38.96.9	46.1996
max	1.07E+13	86.8366	123.1396	123514.1967	9984670	64.9630	178.0650

III. Libraries Used

The library that we used in this project could be separated into different categories:

1. Data Manipulation and Analysis:

a) numpy: For mathematical operations, numerical computations, and handling arrays/matrices.
b) pandas: For data manipulation and analysis, particularly with DataFrames and Series.
c) pycountry_convert: For geographic conversions, such as mapping country codes to regions or continents.

2. Data Visualization:

a) matplotlib.pyplot: For creating 2D plots and visualizing data.
b) seaborn: For statistical data visualization, with advanced integration with pandas.
c) plotly.graph_objects: For building interactive and highly customizable plots.
d) plotly.express: For quickly generating interactive visualizations from tabular data.

3. Data Preprocessing:

a) sklearn.preprocessing.normalize: For normalizing data to scale values uniformly.
b) sklearn.preprocessing.StandardScaler: For standardizing data by removing the mean and scaling to unit variance.
c) sklearn.preprocessing.LabelEncoder: For encoding categorical labels into numerical values.

4. Statistical Analysis:

a) scipy.stats.chi2_contingency: For performing chi-squared tests for statistical independence.

5. Clustering:

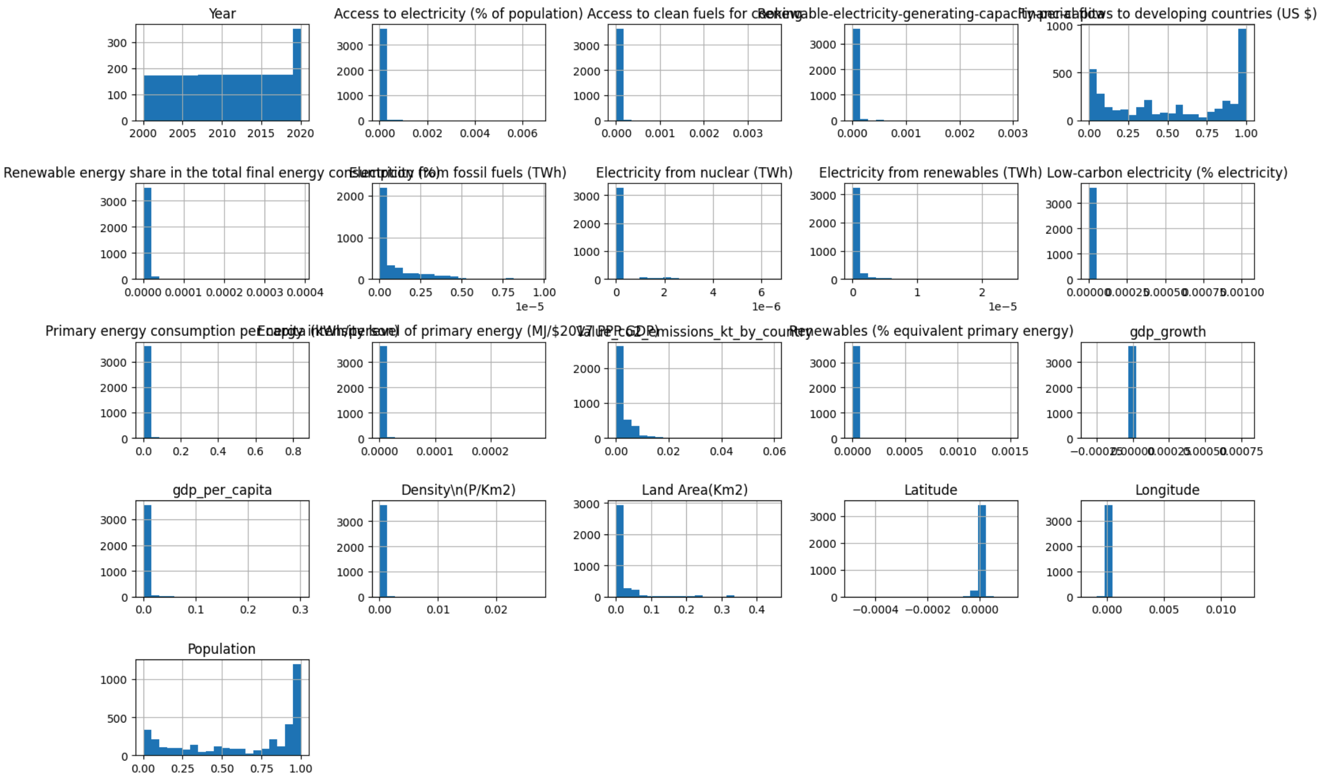
a) DBSCAN: For density-based spatial clustering of applications with noise.

IV. Exploratory Data Analysis

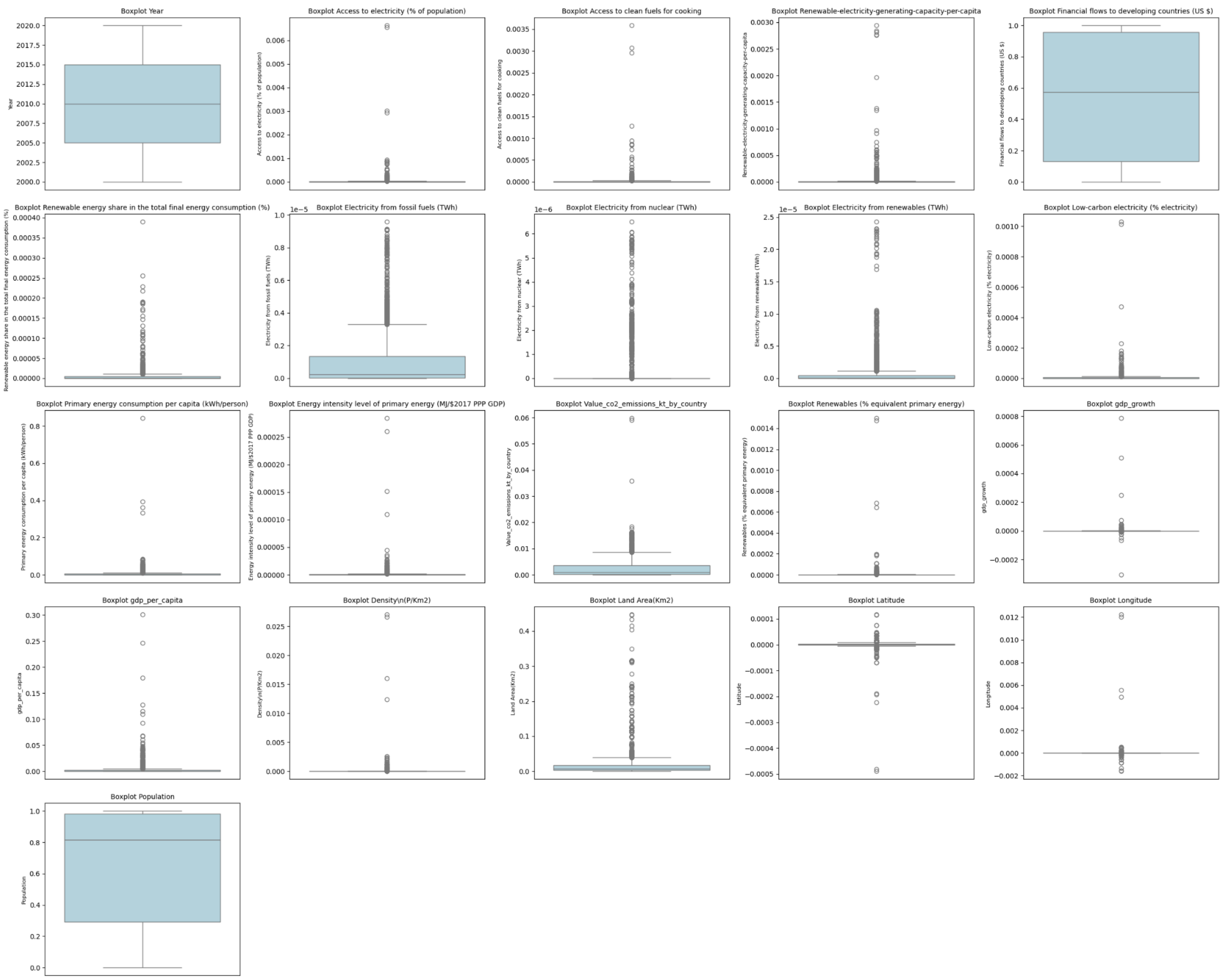
The exploratory data analysis is essential for any data project, because it provides useful information about how data is distributed, it gives insights about features, outliers and trends that data follows. Understanding the structure and the potential issues within the dataset makes applying any ML models much easier.

For this project, we started by implementing a world map with a slider for year to see the evolution of the consumption of different types of electricity such as: Electricity from fossil fuels (TWh), Electricity from nuclear (TWh), Electricity from renewables (TWh), Value_co2_emissions_kt_by_country and also for the GDP. We defined a function called "plot_world_map" and the main library we used is Plotly. This library allowed us to create interactive visualizations, including choropleth maps, which represent data by coloring countries based on specific values. The function filters the dataset for each year, creates a map trace for the given metric, and adds it to the overall figure. By incorporating a slider, we could seamlessly transition through the years from 2000 to 2020, observing trends and changes that happened on a global scale.

Another important step in the exploratory analysis is the visualization of the data. To get different types of visualization we used histograms (for the numerical columns by using the matplotlib library) which provides insights into the distribution of each numerical feature, highlighting patterns, skewness, and potential outliers . Another type of visualization we used are scatter plots to visualize the relationship between each numerical feature and its index, but also to help us identify trends/anomalies/outliers. For those visualizations we used seaborn and matplotlib libraries. The last type of visualization used were boxplots to identify the spread of data and detect outliers for each numerical feature. As in the scatter plots case, we used the same libraries: matplotlib and seaborn. Also, to get the correlations between the columns, we added a correlation matrix by using the corr() function along with seaborn and matplotlib.

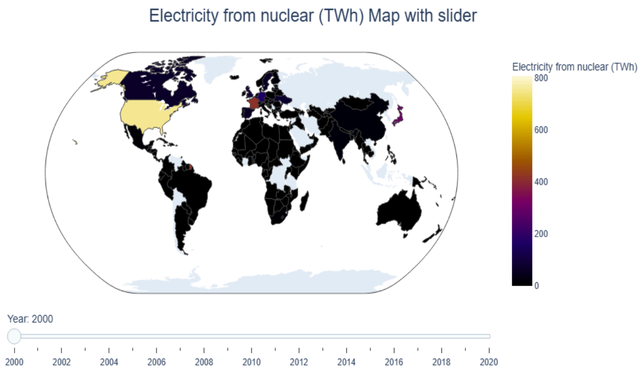


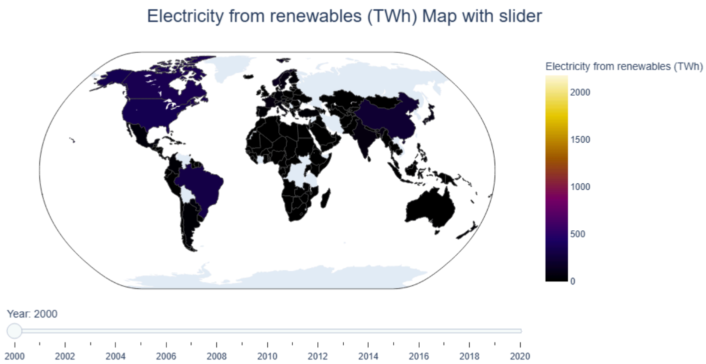
Year
Access to electricity (% of population)
Financial flows to developing countries (US $)
Electricity from fossil fuels (TWh)
Electricity from nuclear (TWh)
Electricity from renewables (TWh)
Low-carbon electricity (% electricity)
Renewables (% equivalent primary energy)
gdp_growth
gdp_per_capita
Density\n(P/Km2)
Land Area(Km2)
Latitude
Longitude
Population

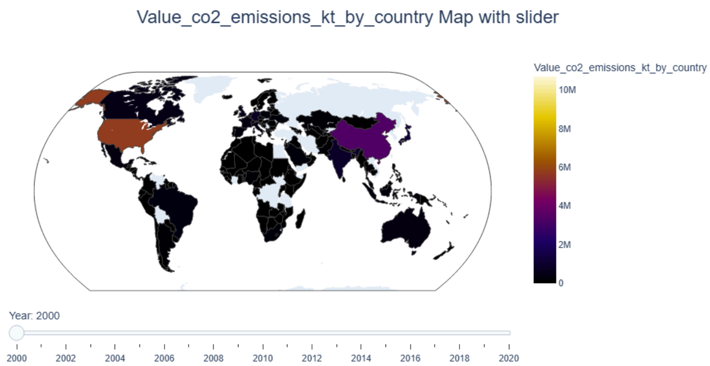


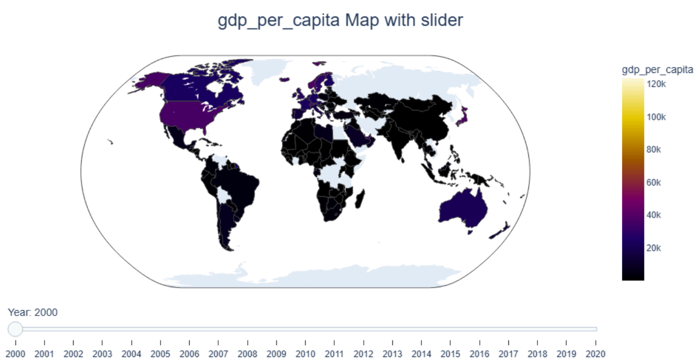
Boxplot Year
Boxplot Access to electricity (% of population)
Boxplot Access to clean fuels for cooking
Boxplot Renewable-electricity-generating-capacity-per-capita
Boxplot Financial flows to developing countries (US $)
Boxplot Renewable energy share in the total final energy consumption (%)
Boxplot Electricity from fossil fuels (TWh)
Boxplot Electricity from nuclear (TWh)
Boxplot Electricity from renewables (TWh)
Boxplot Low-carbon electricity (% electricity)
Boxplot Primary energy consumption per capita (kWh/person)
Boxplot Energy intensity level of primary energy (MJ/$2017 PPP GDP)
Boxplot Value_co2_emissions_kt_by_country
Boxplot Renewables (% equivalent primary energy)
Boxplot gdp_growth
Boxplot gdp_per_capita
Boxplot Density\n(P/Km2)
Boxplot Land Area(Km2)
Boxplot Latitude
Boxplot Longitude
Boxplot Population

These world plots can be visualised with the help of a slider in the notebook.









V. Preprocessing

The next step we took in the project was to do some preprocess and data engineering. The first thing we had to do was to map each country to the right continent, so we could get relevant predictions/classifications. The library we had to use was "pycountry_convert". We defined a function that converts country names to continent codes using the library and assigning them to the continents. In the eventual case when the country code is not mapped to a continent, the function assigns an "Unknown" value.

Since there were some missing values among the data, we removed French Guiana (it only had data for the year 2000) and we also removed the following columns:

- Renewable-electricity-generating-capacity-per-capita - 25.513839% missing values
- Financial flows to developing countries (US $) - 57.248561% missing values
- Renewables (% equivalent primary energy) - 58.563990% missing values

For the remaining columns, we filled the missing values with the median value of the respective column, grouped by the continent to which each country belongs. Also, we standardized and normalized the data using "StandardScaler" and "normalize" functions from sklearn.preprocessing The "L2" method has been used for the normalization, making the data to be between 0 and 1. Another important feature that we added was a population column, which has been calculated by multiplying "Density" and "Land Area". We added this column to have a better understanding for the consumption of energy per capita and also to get better models based on "per capita" columns.

After completing the preprocessing and feature engineering steps, the dataset has been prepared for modeling by splitting it into training and testing subsets by using the function train_test_split from sklearn.model_selection. By doing this step, we ensured that the amount of data used for training was 80% of the total amount, and for testing 20%.

VI. Methods

A. Classification

For the classification task, we used two machine learning algorithms: XGBClassifier and RandomForestClassifier. Both methods were selected for their complementary strengths, with XGBClassifier excelling in feature importance analysis and RandomForestClassifier offering resilience to noisy or incomplete data.

XGBClassifier is a gradient-boosting algorithm that constructs decision trees sequentially, with each tree aiming to correct the classification errors of its predecessors. This iterative process minimizes the loss function and enhances predictive accuracy. The algorithm is particularly effective for feature-rich datasets due to its ability to handle complex interactions between features. Additionally, its computational efficiency and scalability make it suitable for large datasets.

RandomForestClassifier is an ensemble learning method that generates multiple decision trees using random subsets of the data and features. The final classification is determined by averaging the predictions of these individual trees, which enhances robustness and reduces the risk of overfitting. This method is well-suited for datasets with missing values and provides reliable performance across a wide range of classification problems.

B. Clustering

Clustering is a machine learning technique that groups similar data points together based on their features or characteristics. It helps uncover hidden patterns and structures within data without requiring predefined labels. We used DBSCAN for clustering 4 dimensions of data.

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a widely used clustering algorithm in machine learning that groups data points based on their density. It identifies clusters by connecting points that are closely packed and marks points in sparse areas as outliers. Unlike other methods, DBSCAN does not require specifying the number of clusters beforehand and can detect clusters of any shape and size.

C. Regression

Regression models are fundamental tools in machine learning and statistics, allowing us to establish relationships between dependent variables (target) and independent variables (features). Below, we outline the models used, explaining their core characteristics and why they were selected for our study.

Linear Regression is a simple yet powerful baseline model that assumes a linear relationship between the independent variable(s) (X) and the dependent variable (y). It is easy to implement and interpret. Despite its simplicity, it serves as an excellent starting point for evaluating relationships in datasets.

Random Forest is an ensemble learning method that builds multiple decision trees during training and merges their outputs for robust predictions. It excels at handling non-linear relationships and complex datasets, making it effective in capturing intricate patterns. Additionally, it is resilient to outliers and irrelevant features, contributing to its versatility and reliability.

XGBoost (Extreme Gradient Boosting) is a highly efficient and powerful gradient boosting algorithm designed for performance and scalability. It effectively handles non-linear relationships and includes advanced features to mitigate overfitting. XGBoost's ability to optimize computational resources while delivering high accuracy makes it a top choice for large-scale and complex datasets.

Gradient Boosting is a machine learning technique that combines multiple weak learners—typically decision trees—to build a strong predictive model. By sequentially adding trees that correct the errors of the previous models, it minimizes the loss function over iterations. However, it can be slower on large datasets compared to modern variants like XGBoost or LightGBM, making it more suited to smaller-scale problems or where computational efficiency is less critical.

VII. Experiments and Results

A. Continent Classification

For the classification of the continents we firstly defined a function to split the data like mentioned before (80%-train, 20%-test) and for the columns we used to classify the continents we used 'Electricity from fossil fuels (TWh)', 'Value_co2_emissions_kt_by_country', 'Electricity from renewables (TWh)', 'Electricity from nuclear (TWh)', 'gdp_per_capita'. We also defined another function to measure the metrics of the models (accuracy, classification report, confusion matrix) and another one to predict the continents. The models we used for this task were: XGBoosting, RandomForestClassifier and KNeighborsClassifier.

The first model we tested was the **Random Forest** and to get the best parameters for the model, we implemented a Grid Search using the GridSearchCv function from the sklearn.model_selection library. The parameters we tested were the following:

n_estimators	100	**200**	500	
max_depth	10	**20**	30	None
min_samples_split	**2**	5	10	
min_samples_leaf	**1**	5	10	

The best results were obtained by using the bold and underscored values, getting an overall accuracy of 0.90273.

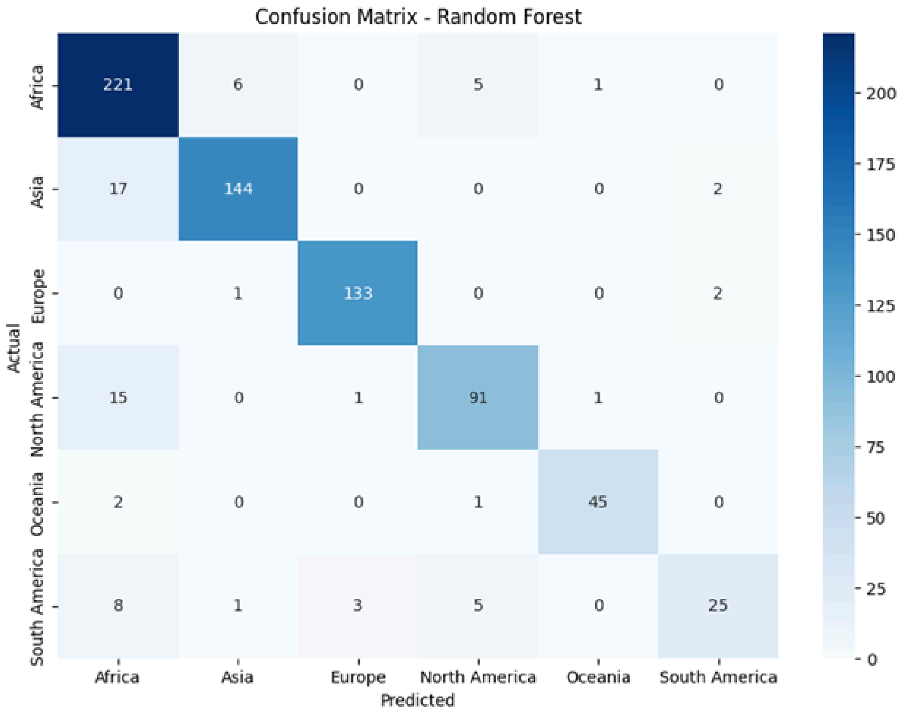
Continent	Precision	Recall	F1-Score	Support
Africa	0.84	0.95	0.89	233
Asia	0.95	0.88	0.91	163
Europe	0.97	0.98	0.97	136
North America	0.89	0.84	0.87	108
Oceania	0.96	0.94	0.95	48
South America	0.86	0.60	0.70	42

Real	Predicted
Asia	Asia

South America	South America
South America	North America
Oceania	Oceania
North America	North America
Asia	Asia
Asia	Asia
South America	South America
Asia	Asia
Europe	Europe

Overall Metrics	**Precision**	**Recall**	**F1-Score**	**Support**
Accuracy	-	-	0.90	730
Macro Avg	0.91	0.86	0.88	730
Weighted Avg	0.91	0.90	0.90	730

Also, the confusion matrix for this model showed us that the model performed quite well.



The second model for the classification task was the XGBoosting classifier. Like in the previous case, we firstly did a Grid Search to get the best parameters to use in the model. The tested values for the parameters were:

n_estimators	190	200	**220**
max_depth	10	**11**	12
learning_rate	0.1	**0.2**	0.3
min_child_weight	**1**	2	10

And obtaining an overall accuracy of 0.882191. The other results were:

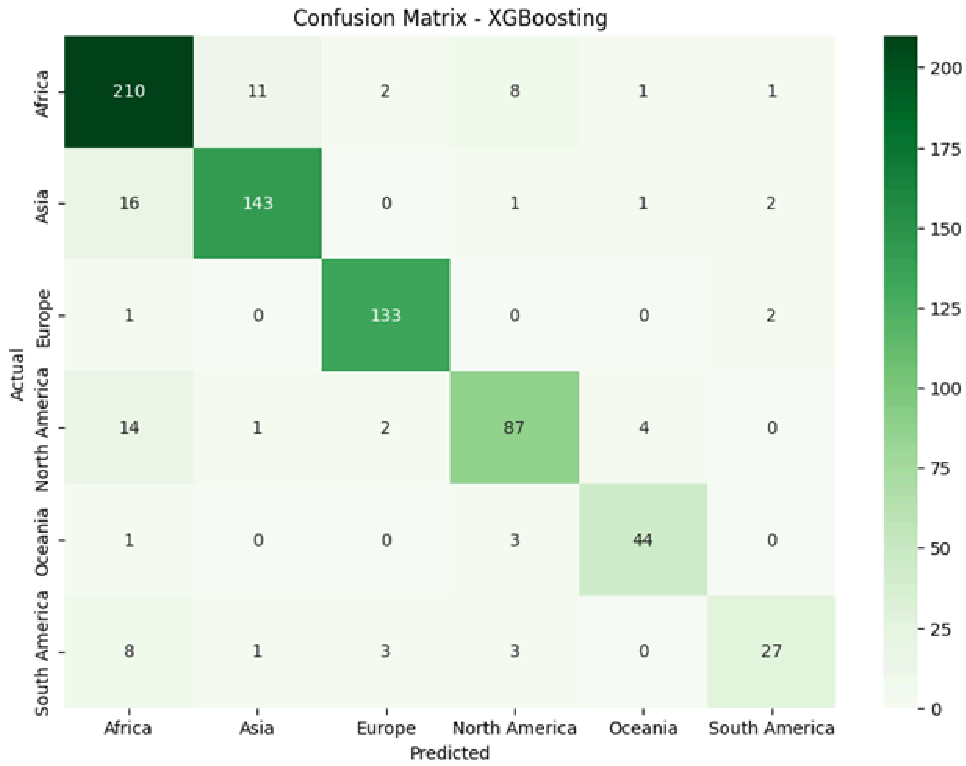
Continent	**Precision**	**Recall**	**F1-Score**	**Support**
Africa	0.84	0.90	0.87	233
Asia	0.92	0.88	0.90	163
Europe	0.95	0.98	0.96	136

North America	0.85	0.81	0.83	108
Oceania	0.88	0.92	0.90	48
South America	0.84	0.64	0.73	42

Real	**Predicted**
Asia	Asia
South America	South America
South America	Europe
Oceania	Oceania
North America	North America
Asia	Asia
Asia	Asia
South America	South America
Asia	Asia
Europe	Europe

Overall Metrics	**Precision**	**Recall**	**F1-Score**	**Support**
Accuracy	-	-	0.88	730
Macro Avg	0.88	0.85	0.86	730
Weighted Avg	0.88	0.88	0.88	730

The confusion matrix for this model looks good, but slightly worse than the Random Forest model:



The last model used for this task was the KNNeighborsClassifier. Like in the other 2 cases, we implemented a grid search to get the best parameters:

N_neighbors	3	5	7	**9**
Weights	Uniform	**Distance**		
Metric	Euclidian	**Manhattan**		
Algorithm	**Auto**	Ball_tree	Kd_tree	brute
Leaf_size	**5**	10	20	30

And getting the worst value of overall accuracy: 0.642465. The other metrics we obtained were:

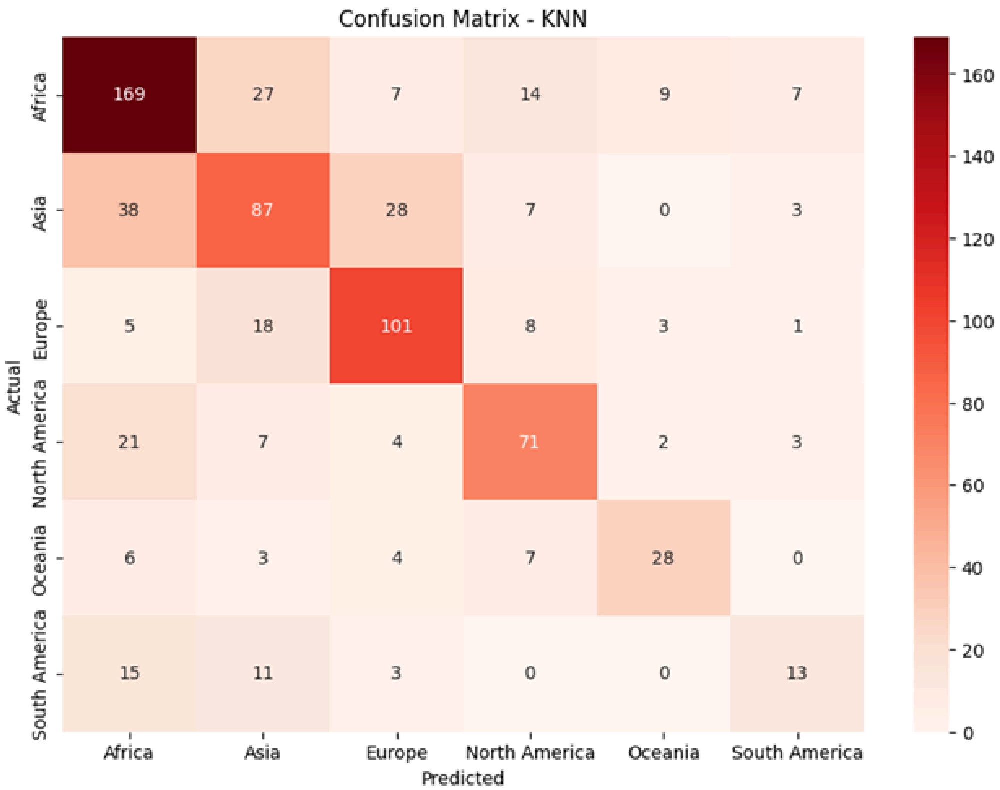
Continent	Precision	Recall	F1-Score	Support
Africa	0.67	0.73	0.69	233
Asia	0.57	0.53	0.55	163
Europe	0.69	0.74	0.71	136

North America	0.66	0.66	0.66	108
Oceania	0.67	0.58	0.62	48
South America	0.48	0.31	0.38	42

Real	**Predicted**
Asia	Asia
South America	Africa
South America	Africa
Oceania	Oceania
North America	North America
Asia	Africa
Asia	Asia
South America	South America
Asia	Europe
Europe	Europe

Overall Metrics	**Precision**	**Recall**	**F1-Score**	**Support**
Accuracy	-	-	0.64	730
Macro Avg	0.62	0.59	0.60	730
Weighted Avg	0.64	0.64	0.64	730

Having the worst accuracy of all 3 models, the KNN had a worse distribution of the continents, as shown in the confusion matrix.

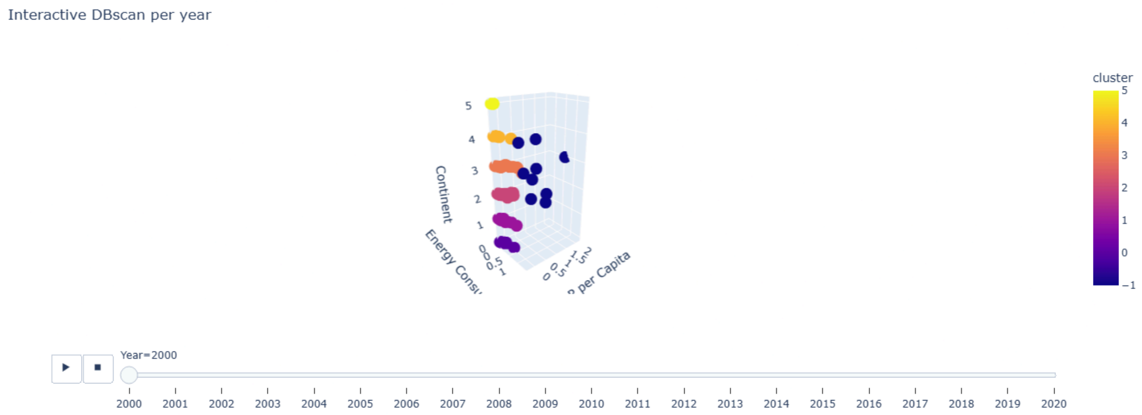


The Random Forest Classifier outperformed the other models, achieving the highest accuracy at 90.27%. This success can be attributed to its ensemble approach, which leverages multiple decision trees to effectively handle non-linear relationships in the data and reduce overfitting. The confusion matrix highlights strong performance in classifying continents like Europe and Oceania, where F1-scores were above 0.95. However, the model faced challenges with South America, where the F1-score dropped to 0.70, suggesting that the features used might not fully capture the unique characteristics of this region.

In contrast, the KNN model performed the worst, with an accuracy of 64.25%. This lower performance is likely due to its reliance on distance metrics, which can become less effective in high-dimensional data like ours. Misclassifications were more frequent for continents such as South America and Asia, as seen in the confusion matrix. This suggests that KNN might not be the best choice for this dataset without additional feature engineering or dimensionality reduction techniques.

B. Clustering of GDP and Power consumption

These world plots can be visualised with the help of a slider in the notebook.



Performance metrics:

Year	Silhouette Score	Number of Cluster	Number of Noise Points
2000	0,79689	7	9
2001	0,78954	7	8
2002	0,77801	7	9
2003	0,76438	7	9
2004	0,75246	7	11
2005	0,75861	7	7
2006	0,75121	7	11
2007	0,73899	7	12
2008	0,73541	7	13
2009	0,72780	7	12
2010	0,74643	7	9
2011	0,70569	7	10
2012	0,73481	7	10
2013	0,72477	7	14
2014	0,67364	7	16
2015	0,71800	7	16

2016	0,73469	7	14
2017	0,71780	7	16
2018	0,71352	7	15
2019	0,71949	7	15
2020	0,76695	7	7

C. GDP Prediction

One of the regression tasks tackled was predicting the **GDP per Capita** in every continent based on **Primary energy consumption**, **Access to clean fuels**, **Access to electricity** and **Electricity generated from renewables**. In order to prepare the data for this task, we separated it into smaller datasets based on the continent from which they are a part of in order to have better and more significant results, given that there are vastly different economic trends continent-wise. For each continent, data was split on a 80%-20% distribution for training and testing. The models compared on this task were **Linear Regression**, **Random Forest** and **XGBoost**.

Linear Regression Results

Continent	**MSE**	**R^2**	**Adjusted R^2**
Africa	0.00000017	0.814179	0.810517
Asia	0.00000023	0.774532	0.768860
Europe	0.00000007	0.671661	0.662787
North America	0.00005148	0.520502	0.502906
Oceania	0.00006274	0.916473	0.909210
South America	0.00000025	0.943414	0.937457

Random Forest Results

Continent	**MSE**	**MAE**	**R^2**	**Adjusted R^2**
Africa	0.00000002	0.00006	0.967804	0.967169

Asia	0.00000014	0.00015	0.858970	0.855422
Europe	0.00000381	0.00077	0.839561	0.835225
North America	0.00004158	0.00198	0.612710	0.598497
Oceania	0.00002584	0.00151	0.965595	0.962603
South America	0.00000021	0.00016	0.953816	0.948954

XGBoost Results

Continent	**MSE**	**MAE**	**R²**	**Adjusted R²**
Africa	0.00000007	0.00012	0.915349	0.913681
Asia	0.00000010	0.00017	0.904966	0.902575
Europe	0.00000182	0.00065	0.923363	0.921292
North America	0.00000687	0.00072	0.936009	0.933661
Oceania	0.00021511	0.00341	0.713647	0.688747
South America	0.00000059	0.00027	0.868872	0.855069

Linear regression's R^2 values range from about 0.52 (North America) to 0.94 (South America). While it does well in Oceania and South America (both > 0.90), it substantially underperforms the tree‑based models for most other continents. The big variations could be due to outliers, data size, or certain continent‑specific features.

Tree‑based models (XGBoost and Random Forest) generally produce lower errors (MSE/MAE) and higher R^2 values than Linear Regression. This suggests that the non‑linear relationships in the data are better captured by Random Forest and XGBoost. Performance varies by continent. Some continents have systematically higher or lower R^2 across all models, possibly reflecting data quality, sample size, or how well GDP scales with the chosen predictors in each region.

XGBoost performs exceptionally well for Asia, Europe, and North America, with R^2 values above 0.90 for each (0.90–0.94 range). Random Forest shows very strong performance in Africa, Oceania, and South America, achieving R^2 values of 0.95–0.97 for those continents. This contrast suggests that no single tree‑based method dominates across all continents; each algorithm may capture different patterns better depending on the region's data characteristics.

D. CO2 Emission Prediction

An important aspect of this project consists in predicting the CO2 values among the continents. To achieve this, we split the data set using the same proportions as in the classification task. The features we used for predicting CO2 values were selected based on their potential correlation with emissions. These features include Electricity from fossil fuels (TWh), GDP per capita, Electricity from renewables (TWh), and Population, while assigning the value to be predicted to the Value_co2_emissions_kt_by_country. The data has been splitted in 80% training and 20% testing, like in the classification task.

To get those predictions we tested out 3 different models: Linear Regression, Random Forest Regressor, and Extreme Gradient Boosting.

The first model we tried was the Linear Regression, for which we obtained the following results

Continent	**MSE**	**R^2**	**Adjusted R^2**	**MAE**
Africa	1.699627e-07	0.904898	0.903024	0.000230
Asia	1.259797e-06	0.836202	0.832081	0.000736
Europe	2.283868e-06	0.661077	0.651917	0.000993
North America	2.062960e-06	0.876799	0.872277	0.001206
Oceania	1.837503e-06	0.935602	0.930002	0.000674
South America	5.337855e-07	0.501075	0.448556	0.000462

The second model we tried was Random Forest Regressor, for which we did a grid search to get the best parameters.

Parameter	**Values**
n_estimators	100, 200, **500**
max_depth	5, 15, 10, 20, **None**

min_samples_split	**2**, 5, 10
min_samples_leaf	**1**, 2, 4
max_features	'auto', **'sqrt'**, 'log2'
bootstrap	**True**, False

For the highlighted values of the parameters, we managed to obtain better performance then in the Linear Regression model:

Continent	MSE	R^2	Adjusted R^2	MAE
Africa	1.264582e-07	0.929241	0.927847	0.000130
Asia	6.715944e-07	0.912680	0.910483	0.000309
Europe	1.452728e-06	0.784417	0.778590	0.000398
North America	1.007679e-06	0.939821	0.937612	0.000519
Oceania	1.028386e-06	0.963959	0.960825	0.000353
South America	6.340918e-07	0.407319	0.344932	0.000369

The last model we used for this task was the XGBoost Regressor. Like in the Random Forest model, we firstly did a grid search with the following parameters:

Parameter	Values
n_estimators	**100**, 200, 500
max_depth	3, 5, **10**, 15
learning_rate	0.01, 0.05, **0.1**, 0.2
subsample	0.6, 0.8, **1**
colsample_bytree	0.6, 0.8, **1**

For the highlighted values, we obtained weaker results than in the previous cases:

Continent	MSE	R^2	Adjusted R^2	MAE
Africa	2.482571e-07	0.861089	0.858352	0.000201
Asia	1.256241e-06	0.836665	0.832556	0.000440
Europe	1.630996e-06	0.757962	0.751420	0.000545
North America	2.271985e-06	0.864318	0.859339	0.000722
Oceania	7.975560e-07	0.972048	0.969618	0.000334
South America	5.874508e-07	0.450914	0.393116	0.000378

The results demonstrate varying levels of success for each model across different continents. The Random Forest Regressor consistently outperformed the Linear Regression and XGBoost Regressor models for most continents, achieving higher R^2 and lower MAE values. This indicates that Random Forest was better at capturing the nonlinear relationships in the data.

In Africa, Random Forest achieved an R^2 of 0.929 compared to 0.904 from Linear Regression, reflecting a notable improvement.

Similarly, in Asia and Europe, Random Forest showed better performance in all metrics. However, South America presented challenges for all models, with significantly lower R^2 values (0.407 for Random Forest and 0.501 for Linear Regression). This may suggest that the selected features do not fully explain CO2 emissions in this region. Additional features or region-specific adjustments might be necessary to improve predictions.

E. Fossil Fuel Consumption Prediction

Another objective of this project was to predict the values of “Electricity from fossil fuels (TWh)” for the continents. This analysis is important to understand the global reliance on fossil fuels and the variations in energy production patterns. To achieve this, we followed the same steps as in the prediction of the CO2 emissions: splitting the data in 80% training and 20% testing, while using these columns for predicting the value of “Electricity from fossil fuels (TWh)” : 'Value_co2_emissions_kt_by_country', 'Electricity from renewables

(TWh)', 'Electricity from nuclear (TWh)', 'Land Area(Km2)', but in this case, we only used 2 models: Linear Regression and Random Forest Regressor.

The first model we teste was a Linear Regression as baseline model. The results were the following:

Continent	MSE	R^2	Adjusted R^2	MAE
Africa	5.033732e-14	0.898812	0.896819	1.030995e-07
Asia	4.381478e-13	0.825675	0.821289	3.905643e-07
Europe	6.584556e-13	0.675502	0.666732	5.555777e-07
North America	9.320403e-13	0.748566	0.739339	6.666960e-07
Oceania	8.934612e-13	0.891913	0.882514	3.556300e-07
South America	8.881514e-14	0.582361	0.538399	1.742698e-07

The second model, Random Forest Regressor, has been fine tunned by using a Grid Search. The parameters were the following:

Parameter	Values
n_estimators	100, 200, **500**
max_depth	5, 15, 10, 20, **None**
min_samples_split	**2**, 5, 10
min_samples_leaf	**1**, 2, 4
max_features	'auto', **'sqrt'**, 'log2'
bootstrap	**True**, False

And for those parameters the obtained results were:

Continent	MSE	R^2	Adjusted R^2	MAE
Africa	1.454145e-14	0.970769	0.970193	5.104846e-08
Asia	1.236941e-13	0.950786	0.949548	1.734547e-07

Europe	1.716240e-13	0.915421	0.913135	2.048234e-07
North America	1.468862e-13	0.960375	0.958921	1.595081e-07
Oceania	1.858474e-13	0.977517	0.975562	1.800684e-07
South America	5.953924e-14	0.720026	0.690555	1.228787e-07

The results demonstrate that the Random Forest Regressor outperformed the Linear Regression model across all continents, achieving higher R^2 scores and lower error metrics such as MSE and MAE. This highlights the model's ability to capture complex, non-linear relationships in the data, which are essential for accurately predicting electricity generation from fossil fuels.

Notably, continents like Oceania and Asia achieved the highest R^2 values, indicating that the selected features were strong predictors in these regions. However, for South America, both models struggled to achieve comparable performance, suggesting that additional region-specific features or data might be required to improve accuracy.

F. Energy from Nuclear Sources Prediction Using World and Continental Data

Model Comparisons and Performance Metrics for World Data

To predict energy trends and their relationship with key variables, two machine learning models, Gradient Boosting Regressor and Random Forest Regressor, were employed. The models were evaluated using global data and specific features, with their performance assessed through RMSE (Root Mean Square Error) and R^2 (coefficient of determination) scores.

Gradient Boosting Regressor:

Key Features: Low-carbon electricity (% of electricity), CO_2 emissions (kt) by country, and population.

Results: Achieved an RMSE of 2.20×10^{-7} and an R^2 of 0.93, indicating strong predictive accuracy and a robust ability to explain the variance in the data.

Random Forest Regressor:

Key Features: Electricity from fossil fuels (TWh), CO_2 emissions (kt) by country, and population.

Results: Delivered a lower RMSE of 1.35×10^{-7} and a higher R^2 of 0.97, highlighting its superior performance in capturing the variability of the data and producing more precise predictions.

	RandomForestRegressor			**GradientBoostingRegressor**		
Continent	RMSE	R2	MAE	RMSE	R2	MAE
Africa	2.575708e-08	0.212639	5.309925e-09	2.426469e-08	0.301237	4.820007e-09
Europe	3.041500e-07	0.964693	1.684105e-07	2.983686e-07	0.966023	1.861199e-07
South America	1.335337e-08	0.651589	7.067470e-09	1.139766e-08	0.746171	6.330422e-09
Asia	1.759961e-07	0.447525	3.774996e-08	1.284919e-07	0.705518	3.602300e-08
North America	1.044513e-07	0.975822	2.778816e-08	9.495151e-08	0.980020	3.145911e-08
Oceania	0.000000e+00	1.000000	0.000000e+00	0.000000e+00	1.000000	0.000000e+00

Regional Insights from Per-Continent Data

The models' performance was also analyzed at the continental level, revealing diverse predictive accuracy across regions:

Africa: Both models struggled to explain energy patterns effectively, with R^2 values below 0.30, suggesting the presence of more complex, unaccounted factors influencing energy trends.

Europe: Exhibited high predictive accuracy, with R^2 values exceeding 0.96, reflecting a more stable relationship between the chosen features and energy outcomes.

South America: Moderate predictive success was observed, with R^2 values around 0.65–0.75, indicating partial alignment between the models' features and energy trends.

Asia: Performance was mixed, with R^2 values ranging from 0.45 to 0.70. This suggests substantial variability in energy trends and the need for more tailored features to improve predictions.

North America: The models performed exceptionally well, achieving R^2 values above 0.97, highlighting strong alignment between the input features and energy patterns in this region.

Oceania: Both models achieved perfect predictions, with R^2 values of 1.0, likely due to the smaller size and homogeneity of the dataset for this region.

Key Findings

Global Models: Random Forest Regressor outperformed Gradient Boosting Regressor globally, achieving both a lower RMSE and a higher R^2 score.

Regional Variations: The predictive accuracy of both models varied significantly across continents, with Europe, North America, and Oceania demonstrating strong results, while Africa and Asia exhibited lower R^2 scores, indicating the need for additional or alternative features to capture regional energy patterns.

Feature Impact: The inclusion of variables such as population, CO_2 emissions, and electricity generation sources (e.g., fossil fuels vs. low-carbon) proved effective, though their relative influence differed between regions and models.

To determine the optimal feature set for both the RandomForestRegressor and the GradientBoostingRegressor models, I employed the Recursive Feature Elimination (RFE) technique. By systematically removing the least important features and re-evaluating the model at each step, RFE iteratively narrows down the feature set until only the most critical ones remain. For this analysis, the RFE selector was configured with n_features_to_select=3, meaning the algorithm was tasked with identifying the top three features that contributed the most to the predictive accuracy of the models.

G. Energy from Renewable Sources Prediction Using World and Continental Data

Model Comparisons and Performance Metrics for World Data

The prediction models for energy derived from nuclear sources focused on key variables such as low-carbon electricity percentage, CO_2 emissions by country (in kilotons), and population. Two regression models were utilized:

RandomForestRegressor

Hyperparameters:

n_estimators = 500, max_features = sqrt, bootstrap = True

Performance Metrics:

RMSE: 3.4693442764460045e-07,

R2: 0.9646372719842592

GradientBoostingRegressor

Hyperparameters:

n_estimators = 500, learning_rate = 0.1, max_features = sqrt

Performance Metrics:

RMSE: 1.3457393428352056e-07

R2: 0.9749121622580753

The RandomForest model outperformed Gradient Boosting in terms of lower RMSE and higher R2, indicating its superior ability to handle the non-linear relationships between predictors and target variables in this context.

To predict energy derived from renewable sources, the models incorporated various predictors, including land area (km^2), primary energy consumption per capita (kWh/person), and low-carbon electricity (% electricity) for the RandomForest model, and electricity from fossil fuels (TWh), population, and CO_2 emissions (kt) for GradientBoosting.

	RandomForestRegressor			**GradientBoostingRegressor**		
Continent	RMSE	R2	MAE	RMSE	R2	MAE
Africa	3.534943e-08	0.812180	2.097163e-08	4.428311e-08	0.705250	2.547735e-08
Europe	3.594215e-07	0.983866	2.376551e-07	3.844853e-07	0.981538	2.447170e-07
South America	3.859115e-07	0.964761	2.267763e-07	2.732333e-07	0.982335	1.859305e-07
Asia	1.149164e-07	0.907081	6.432266e-08	1.750042e-07	0.784504	8.254404e-08
North America	1.708109e-07	0.992596	8.124384e-08	1.336881e-07	0.995465	6.384418e-08
Oceania	5.257513e-07	0.928363	1.873588e-07	3.843518e-07	0.961715	1.399104e-07

Regional Analysis of Renewable Energy Prediction

The models were further evaluated on a per-continent basis to understand their performance in regional contexts. The following observations summarize the results:

Africa: RandomForest yielded higher accuracy (R2=0.812) compared to GradientBoosting (R2=0.705).

Europe: Both models performed exceptionally well.

Asia: GradientBoosting outperformed RandomForest.

North America: Both models demonstrated exceptional performance, with R2 values close to 0.99, but GradientBoosting exhibited slightly lower RMSE, making it the more accurate predictor.

Oceania: The models had good performance, with Gradient Boosting giving slightly better results.

In summary, both models exhibited strengths across continents, but their performance varied based on the complexity and features of regional data.

To determine the optimal feature set for both the RandomForestRegressor and the GradientBoostingRegressor models, I employed the Recursive Feature Elimination (RFE) technique. By systematically removing the least important features and re-evaluating the model at each step, RFE iteratively narrows down the feature set until only the most critical ones remain. For this analysis, the RFE selector was configured with n_features_to_select=3, meaning the algorithm was tasked with identifying the top three features that contributed the most to the predictive accuracy of the models.

VIII. Conclusions

Access to energy remains a cornerstone for fostering development and improving well-being, underscoring the need for a sustainable transition to renewable energy sources.

Between 2000 and 2020, we analyzed global energy and development data using eight models: four regression, three classification, and one DBSCAN clustering. Our findings highlight renewable energy's vital role in reducing fossil fuel reliance, the high correlation between wealthy nations and energy consumption and the low adoption of renewable energy by the poorest countries, with Random Forest and XGBoost outperforming other methods.

Performance varied by continent, underscoring the need for region-specific approaches, especially in South America and Africa. Effective preprocessing (ex, normalization, feature selection) and novel metrics (population, per capita data) improved model accuracy. Overall, these results stress the importance of data-driven strategies and clean energy investments to address global energy challenges and climate change.

IX. Resources

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